Eloy Alvarado Narváez

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Employment

2026 –	Assistant Professor, Pontificia Universidad Católica de Chile	*Santiago, Chile.*
2023 – 2025	Academic Instructor, Universidad Técnica Federico Santa María	*Santiago, Chile.*
2022 – 2022	Lecturer, Universidad Técnica Federico Santa María	*Santiago, Chile.*
2021 – 2022	External Adviser, Instituto de Fomento Pesquero	*Valparaíso, Chile.*
2019 – 2022	Lecturer, Universidad de Valparaíso	*Valparaíso, Chile.*
2017 – 2017	Lecturer, Pontificia Universidad Católica de Valparaíso	*Valparaíso,Chile.*
2016 – 2017	Statistical Analyst, Equifax Inc.	*Santiago, Chile. / Atlanta, US.*

Education

2018 – 2022	Ph.D. in Statistics, Instituto de Estadística, Universidad de Valparaíso Advisors: Moreno Bevilacqua and Christian Caamaño	*Valparaíso, Chile.*
2013 – 2015	B.Sc in Statistical Science, Pontificia Universidad Católica de Valparaíso	*Valparaíso, Chile.*
2014	Exchange program in Korea University	*Seoul, South Korea*

Research Articles

1. H. Araya, F. Plaza-Vega, and E. Alvarado. "Drift parameter identification for the Ornstein-Uhlenbeck process driven by Ornstein-Uhlenbeck with small General Gaussian noise". In: *Statistical Papers* (2025). Accepted for publication.

2. E. Alvarado, F. Plaza-Vega, C. Montenegro, and O. Saavedra. "Deep learning-based classification of species in central-southern fisheries in Chile". In: *Latin American Journal of Aquatic Research* 53.3 (2025), pp. 411–424. ISSN: 0718-560X. DOI: 10.3856/vol53-issue3-fulltext-3318.

3. M. Bevilacqua, E. Alvarado, and C. Caamaño-Carrillo. "A flexible Clayton-like spatial copula with application to bounded support data". In: *Journal of Multivariate Analysis* 201 (2024). Copula Modeling from Abe Sklar to the present day, p. 105277. ISSN: 0047-259X. DOI: https://doi.org/10.1016/j.jmva.2023.105277.

4. V. Beltrán, M. Flores, C. Sanzana, F. Muñoz-Sepúlveda, E. Alvarado, B. Venegas, J. C. Molina, S. Rueda-Velásquez, and A. von Marttens. "Tooth Loss and Caries Experience of Elderly Chileans in the Context of the COVID-19 Pandemic in Five Regions of Chile". en. In: *International Journal of Environmental Research and Public Health* 20.4 (Jan. 2023), p. 3001. ISSN: 1660-4601. DOI: 10.3390/ijerph20043001. (Visited on 02/25/2023).

5. V. Beltrán, P. A. Mardones, J. Díaz, E. Alvarado, and A. von Marttens. "TEGO: A new concept of teledentistry for the elderly through a web platform and mobile app in the context of the covid-19 pandemic." In: *Journal of Oral Research* 1 (2022), pp. 1–8. DOI: 10.17126/joralres.2022.023.

6. V. Beltrán, A. von Marttens, P. Acuña-Mardones, C. Sanzana-Luengo, S. J. Rueda-Velásquez, E. Alvarado, M. Flores, A. Cerda, and B. Venegas. "Implementation of a Teledentistry Platform for Dental Emergencies for the Elderly in the Context of the COVID-19 Pandemic in Chile". In: *BioMed Research International* 2022 (2022). DOI: 10.1155/2022/6889285.

7. C. Sanzana-Luengo, L. Díaz, L. Abarza, E. Alvarado, and V. Beltrán. "Teledentistry protocol for the elderly in the context of the COVID-19 pandemic through a web platform/mobile app: approach from the general dentist." In: *Journal of Oral Research* (2022), pp. 1–8. DOI: 10.17126/joralres.2022.026.
8. M. Stehlík, J. Kisel'ák, A. Dinamarca, E. Alvarado, F. Plaza, F. Medina, S. Stehlíková, J. Marek, B. Venegas, A. Gajdoš, et al. "REDACS: Regional emergency-driven adaptive cluster sampling for effective COVID-19 management". In: *Stochastic Analysis and Applications* (2022), pp. 1–35. DOI: 10.1080/07362994.2022.2033126.

Under Review

1. E. Farías, E. Alvarado, and J. Scavia. *Contagion, Financial Distress and Divergent Beliefs.*
2. H. Araya, F. Plaza-Vega, and E. Alvarado. *Drift parameter estimation for the noisy Ornstein-Uhlenbeck process driven by a Rosenblatt process.*

Working Papers

1. D. Godoy, E. Alvarado, and T. Chávez. *From LiDAR Intensity to Smart Maintenance: A Weibull Random-Field Framework for Diagnosis, Prognosis, and Decision Automation on Mining Haul Roads.*
2. M. Bevilacqua, E. Alvarado, and C. Caamaño-Carrillo. *Ali-Mikhail-Haq spatial copula.*
3. E. Alvarado, C. Caamaño-Carrillo, and D. Godoy. *Weibull random fields through Clayton spatial copula: An application to mining haul roads.*
4. F. Días, E. Alvarado, and V. Cruz. *Non-Standard Profit Efficiency in Banking: A copula approach.*
5. E. Alvarado, M.E. Farías, K. Martínez, and A. Alfaro. *A CAPM Approach with Stable Error Distributions: Application to Brazilian Market Dynamics.*
6. E. Alvarado, M. Bevilacqua, and C. Caamaño-Carrillo. *Gumbel Spatial Copula.*

Grants and Projects

2025 – 2027 Spatial extension of the Gumbel copula: applications to some non-Gaussian random fields, Fondecyt de Iniciación 2025 ANID No. 11250479.

2023 – 2024 Modeling and estimating massive point referenced spatial data, MATH-AMSUD AMSUD220041. Research Associate.

2018 – 2022 National Doctoral Degree, ANID No. 2018-21180953.

2020 – 2020 Semi-presential technological support platform for urgent and priority dental care for the elderly in the context of the COVID-19 pandemic in the Chilean population, ANID, Reference No. COVID0766, Chile. Collaborator.

Teaching

Pontificia Universidad Católica de Chile

EYP2307	Regression Analysis.	*Taught in 2026/01*

Universidad Técnica Federico Santa María

ICN292	Management Information Systems.	*Taught in 2023/01, 2024/01–02, 2025/01*
PII402	Stochastic Processes (Postgraduate).	*Taught in 2023/02, 2024/01, 2025/01*
IND163C	Business Analytics.	*Taught in 2022/02, 2023/01*

Universidad Andrés Bello

CIND211	Stochastic Networks and Systems.	*Taught in 2025/02*
ATDFI102	Introduction to programming.	*Taught in 2024/01,2025/01*
CIND217	Business intelligence.	*Taught in 2024/01, 2025/01*
IICGP01	Business intelligence.	*Taught in 2023/02*

Universidad de Valparaíso

IEST422	Statistical Quality Control.	*Taught in 2022/02*
LFIS325	Statistics for Physical Sciences.	*Taught in 2022/02, 2021/02, 2020/02*
IECD411	Machine Learning.	*Taught in 2022/01*
IEST414	Time Series.	*Taught in 2022/01*
IECD415	Multivariate Methods.	*Taught in 2022/01*
IECD325	Linear Models and Experimental Design.	*Taught in 2021/02*
IEST412	Sampling Theory II.	*Taught in 2021/01-02, 2020/01*
IEST322	Sampling Theory I.	*Taught in 2021/01, 2020/02, 2019/02*
IEC311	Probability and Statistics.	*Taught in 2021/01, 2020/01*
IEST213	Statistical Methods.	*Taught in 2019/02*

Pontificia Universidad Católica de Valparaíso

EST205	Probability and Statistics.	*Taught in 2017/01-02*

Academic Service

2024 – 2024 Scientific Committee Member, XLVII National Conference on Statistics, organised by the Chilean Statistical Society.

2024 – 2025 Internship Coordinator, Department of Industries, Campus Vitacura, Universidad Técnica Federico Santa María.

2024 – 2025 Member of the academic committee for the development of a new undergraduate program: Bachelor's Degree in Data Science, Universidad Técnica Federico Santa María.

Students

2025 José Donoso. Industrial Engineering. Universidad Técnica Federico Santa María. Advisor.

2025 Wilson Godoy. Industrial Engineering. Universidad Técnica Federico Santa María. Advisor.

2025 Isadora Araya. Industrial Engineering. Universidad Técnica Federico Santa María. Advisor.

2025 Gabriela Estrada. Industrial Engineering. Universidad Técnica Federico Santa María. Advisor.

2025 Fernanda Pincheira. Industrial Engineering. Universidad Técnica Federico Santa María. Advisor.

2025 Benjamín Mardones. Industrial Engineering. Universidad Técnica Federico Santa María. Advisor.

2024 Luis Riquelme. Master in Mathematics.Universidad del Bío-Bío. Co-Advisor.

2024 Cristóbal Retamal. Industrial Engineering. Universidad Técnica Federico Santa María. Advisor.

2024 Nicolás Bravo. Industrial Engineering. Universidad Técnica Federico Santa María. Advisor.

2023 Ricardo Menares. Statistics and Data Science Engineering. Universidad de Valparaíso. Co-advisor.

2022 Cristóbal Collao. Statistical Engineering. Universidad de Valparaíso. Co-advisor.

Scientific Events

Mar 2026 *"Spatial copula models for non-gaussian random fields"*. Contributed session (Spatial processes and applications), 17th Latin American Congress of Probability and Mathematical Statistics (CLAPEM 17), Montevideo. Organizer: CLAPEM 17

Nov 2025 *"Copula-Based Spatial Random Fields: Theory and Applications"*. Universidad del Bío-Bío. Organizer: Universidad del Bío-Bío

Sep 2025 *"Non-Standard Profit Efficiency in Banking: A Copula Approach"*. Campus Vitacura Colloquium. Organizer: Universidad Técnica Federico Santa María

Mar 2025 *"2nd Workshop: Reproducible Research with R"*. Open Science Webinar USM. Organizer: Universidad Técnica Federico Santa María

Nov 2024 *"Weibull random fields through Clayton spatial copula An application to mining haul roads"*. III Conference on Geostatistics and Spatio-Temporal Statistics: Theory and Applications. Organizer:Universidad Técnica Federico Santa María

Nov 2024 *"Weibull random fields through Clayton spatial copula An application to mining haul roads"*. III Statistical Engineering Conference. Organizer: Universidad de Santiago de Chile

Oct 2024 *"Archimedean-like spatial copulas"*. XLVII National Conference on Statistics. Organizer: Chilean Statistical Society

Oct 2024 *"Analysis of the Intention to Use ICT in Industrial Engineering: Integrating UTAUT and Structural Equation Models"*. XXXVI Chilean Congress of Engineering Education. Organizer: Chilean Society of Engineering Education

Aug 2024 *"Workshop: Reproducible Research with R"*. Open Science Webinar USM. Organizer: Universidad Técnica Federico Santa María

Jul 2024 *"Experiences in Open Science"*. Open Science USM. Organizer: Universidad Técnica Federico Santa María

Jul 2024 *"Archimedean-like spatial copulas and their applications"*. Colloquium on Statistics and Data Science. Organizer: Pontificia Universidad Católica de Chile

Nov 2023 *"Tools and Strategies for Reproducible Research"*. 3rd Open Science Webinar USM. Organizer: Universidad Técnica Federico Santa María

Oct 2023 *"AI-based Feedback for Engineering Students"*. XXXV Chilean Congress of Engineering Education. Organizer: Chilean Society of Engineering Education

Dec 2022 *"A flexible Clayton-like spatial copula with application to bounded support data"*. III Statistics Workshop: Graduate Contributions and Scientific initiation. Organizer: Chilean Statistical Society

Oct 2015 *"Stochastics differential equations in mathematical finance: The Black-Scholes Model"*. XXI Valparaíso Statistical Week. Pontificia Universidad Católica de Valparaíso

References

- Moreno Bevilacqua, Associate Professor at Universidad Adolfo Ibañez. `moreno.bevilacqua@uai.cl`
- Christian Caamaño, Associate professor at Universidad del Bío-Bío. `chcaaman@ubiobio.cl`
- Oscar Saavedra, Associate Professor at Universidad Técnica Federico Santa María. `oscar.saavedra@usm.cl`
- Carlos Montenegro, Head of Fisheries Research at IFOP. `carlos.montenegro@ifop.cl`

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